

SOF IMO Cheat Sheet — Class III Math

Conversions

Length	Weight	Capacity	Time	Money
1 m = 100 cm	1 kg = 1000 g	1 L = 1000 mL	1 h = 60 min	₹1 = 100 paise
1 km = 1000 m	half kg = 500 g	half L = 500 mL	half h = 30 min	50 p + 50 p = ₹1
half m = 50 cm			1 day = 24 h; 1 week = 7 days	Change = given – cost
			1 year = 12 months = 365 days	

Big unit → small unit: MULTIPLY. Compare only in the SAME unit (1 kg = 1000 g > 750 g).

Division = tables backwards

- $56 \div 8 \rightarrow$ think " $8 \times ? = 56$ " $\rightarrow 7$. Know tables 2–10 both ways.
- $n \div 1 = n$; $n \div n = 1$; $0 \div n = 0$. Remainder is always SMALLER than the divisor.
- "Boxes needed" $\rightarrow$ round UP; "boxes completely full" $\rightarrow$ round DOWN.

Fraction of a number

- **$1/n$ of a number = number $\div$ n.** $1/2$ of 16 = 8; $1/4$ of 20 = 5.
- $2/3$ of 12 = $(12 \div 3) \times 2 = 8$.
- Unit fractions: bigger bottom = smaller piece $\rightarrow 1/8 < 1/4 < 1/3 < 1/2$.
- "Left" fraction = parts left / total parts (ate 3 of 8 $\rightarrow 5/8$ left).

Shapes

Shape	Sides/Corners	Solid	Faces / Edges / Corners
Triangle	3 / 3	Cube, cuboid	6 / 12 / 8
Square	4 equal / 4	Sphere	0 flat / 0 / 0
Rectangle	4 (opp. equal) / 4	Cylinder	2 flat / 0 / 0
Pentagon, hexagon, octagon	5, 6, 8	Cone	1 flat / 0 / 1 point
Circle	0 / 0		

Pictograph: read the KEY (1 symbol may = 2 or 5). Tally |||| (crossed) = 5.

Clock

- Short hand = hour, long hand = minutes. :15 = quarter past, :30 = half past.
- **:45 = quarter to the NEXT hour** (5:45 = quarter to 6).
- Duration: count on from start to end (4:15 $\rightarrow$ 5:00 = 45 min).